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Melting the driveway with water you have already paid for

Laugarvirki hf. · Geothermal district heating for Reykjavík, Kópavogur, Garðabær and Mosfellsbær · Ref. VAR-2026-31 · Published 1 October 2026

Before the first frost, our network team explains how to size a snow-melt loop on your house's return water, when you need a sealed top-up valve, and what the new cold-alert rule means for your steps and driveway this winter.

A second life for warm water

Every cubic metre of hot water that reaches a Laugarvirki customer arrives at about 76 °C and leaves the radiators at roughly 30 °C. Because you pay by volume at the main meter, the heat still left in that return water costs you nothing extra. Running it through plastic pipes under a driveway, path or front steps before it goes to the drain is the cheapest way to keep paving free of ice in the capital area, and around 41,000 of our 68,500 connected properties already do it.

The catch is that return water alone is often too cool, or there is too little of it, on the coldest mornings. Many homeowners then add a *top-up valve* that mixes in fresh supply water. That water is metered and billed, and on cold-alert days it competes with radiators across the city. This feature shows how to work out whether you need one.

One household's numbers

Halldóra Sveinsdóttir and Einar Bragason live in a 1987 terraced house on Logafold in Grafarvogur. Their 48 m² driveway and 10 m² path stayed icy along the north edge every January, and the six concrete steps to the door (6 m²) were the worst of all. Their plumber measured a winter return flow of 0.40 m³/h at 31 °C. "We assumed we needed a bigger loop," says Halldóra. "In fact the loop was fine; there just wasn't enough warm water going through it." The worked example below uses their figures.

Design melt rates

Network engineer Þórdís Kjartansdóttir sets the planning values in our Snow-Melt Connection Guide, edition 4 (May 2026):

- Residential driveways, paths and parking spaces: **220 W/m²** (the 2019 edition used 250 W/m²).
- Steps, ramps and landings: **320 W/m²**, because water runs off and wind exposure is higher.
- Commercial forecourts and loading bays: 380 W/m².
- Design outlet temperature of the loop: **10 °C**; never plan below 5 °C, as the pipe surface may then freeze.

$$P = q_d A_d + q_s A_s \tag{1}$$

where P is the design melting power of the area (W), q_d the melt rate for driveways, paths and parking (220 W/m²), A_d their area (m²), q_s the melt rate for steps, ramps and landings (320 W/m²) and A_s their area (m²). Divide by 1,000 for kW.

$$P_r = k \cdot F_r \cdot \left(T_r - T_o \right) \tag{2}$$

where P_r is the heat available from the return water (kW), k = 1.15 kWh per m³ per °C the heat content of geothermal water used in all our guides, F_r the winter return flow of the house (m³/h), T_r the return temperature entering the loop (°C) and T_o the loop outlet temperature (°C, normally 10 °C).

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$$F_t = \frac{P - P_r}{k \cdot (T_s - T_o)} \tag{3}$$

where F_t is the top-up flow of fresh supply water needed (m³/h), P the design power from (1) in kW, P_r the return-water heat from (2), and T_s the supply temperature: 76 °C in supply zones A to G, 71 °C in zone H (Úlfarsárdalur and Norðlingaholt). If P is not larger than P_r, no top-up is needed.

$$C = p \cdot F_t \cdot H \tag{4}$$

where C is the extra hot-water cost for one winter (ISK), p the volume price (ISK 167 per m³ from 1 September 2026; ISK 158 until 31 August 2026), F_t the top-up flow from (3) and H the hours per winter the top-up valve is open (we plan with 350 h for the capital area, 420 h for Mosfellsbær above 100 m).

Worked example: Logafold

1. Design power, equation (1): 58 m² × 220 W/m² + 6 m² × 320 W/m² = 12,760 + 1,920 = 14,680 W, i.e. **14.68 kW**.
2. Return-water heat, equation (2): 1.15 × 0.40 × (31 − 10) = **9.66 kW**. The deficit is 14.68 − 9.66 = 5.02 kW.
3. Top-up flow, equation (3), zone C: 5.02 / (1.15 × (76 − 10)) = 0.0661 m³/h, about **66 litres per hour**.
4. Winter cost, equation (4): 0.0661 × 350 h = 23.1 m³, and 23.1 × 167 = about **ISK 3,870** a winter (ISK 3,660 at the old price).

The family's top-up valve was sealed at 66 L/h in March 2026. Einar notes that the steps now clear before the school run.

Registering a top-up valve

A top-up valve may only be fitted and opened after Laugarvirki has registered and sealed it. The order is:

1. Your master plumber, licensed in Iceland, sends form **SB-2** through the Mínar síður customer portal with the calculation above and a sketch of the loop.
2. We review the form within 10 working days and reply with a valve size and flow limit.
3. The plumber fits a thermostatic limiter set to no more than 45 °C at the loop inlet, and a separate top-up meter which we supply (rental ISK 38 per day, billed with your heating account).
4. A Laugarvirki inspector visits, checks the limiter and puts a numbered lead seal on the valve. Only then may it be opened.

The registration fee is ISK 14,900. Breaking a seal, or running an unregistered top-up, is charged ISK 45,000 per occasion under section 9 of our General Terms of Supply, and we may remove the valve.

Cold-alert days

When the forecast for Reykjavík is −8 °C or colder for two or more consecutive days, we declare a *Kuldavakt* (cold alert) by text message and on our website at least 12 hours in advance. During a Kuldavakt, every sealed top-up valve must be closed from 06:00 to 11:00 and from 17:00 to 20:00, so that radiators get the full supply pressure at peak times.

The rule covers top-up water only. A loop that runs purely on the house's own return water may stay on around the clock, because it takes no extra water from the network. The one exception for top-up valves is a ramp or step registered as an accessible entrance (form SB-4) at a care home, health clinic or the home of a registered wheelchair user: these may stay open during peak hours, limited to the ramp or steps only.

Last winter we called four Kuldavakt periods totalling 11 days; on 17 January 2026 peak demand reached 2,140 litres per second.

Before you dig. Loops must be pressure-tested at 6 bar for two hours before paving, and the return line must still pass through the main meter's return side. Our sizing check is free.

Varmi is edited by Bjarki Steinþórsson. Snow-melt questions: Þórdís Kjartansdóttir, Network Engineering, Laugarvirki hf., Viðarhöfði 7, 110 Reykjavík · tel. +354 516 4400 · snjor@laugarvirki.is · Customer service open weekdays 08:30–16:00.

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